## Understanding check

**Question.** For a positive example with predicted probability $p = 0.2$, what is $\partial \ell / \partial z$ for binary cross-entropy, and should gradient descent raise or lower $z$?

**Answer.** $p - y = -0.8$; subtracting a negative gradient raises $z$.

**Misconception diagnosed.** Confusing the gradient sign with the direction of the parameter update.